



Guangdong Technion

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GTIIT

CALCULUS 1 - M

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LECTURE 8 - NOVEMBER 6TH

Velocity is the derivative of the Position

Suppose an object moves according to the position function

$$y = f(t)$$

which gives the displacement of the object from the origin at time t . In the time interval from t to $t + h$, the change in position is $\Delta y = f(t + h) - f(t)$.

The **average velocity** over this time interval is the quotient $\frac{\Delta y}{\Delta t}$

$$= \text{average velocity} = \frac{\text{displacement}}{\text{time}} = \frac{f(t + h) - f(t)}{h}$$

which is the same as the **slope of the secant line**, which intersects the curve at $(t, f(t))$ and $(t + h, f(t + h))$

Now suppose we compute the average velocities over shorter and shorter time intervals $[t, t + h]$

⇒ we let h approach 0

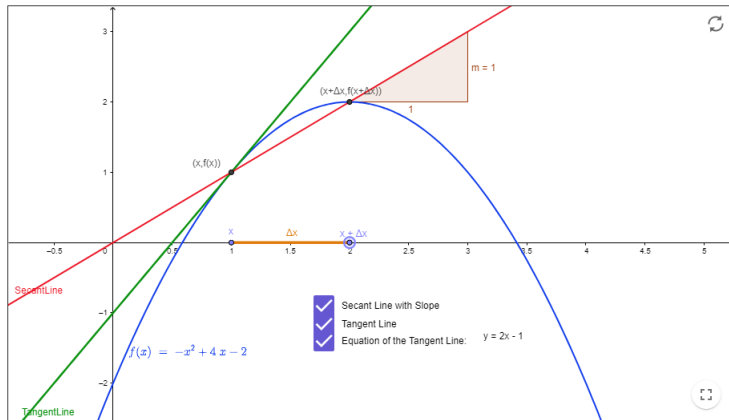
⇒ the **instantaneous velocity** at the time t is so the limit of these average velocities:

$$\text{instantaneous velocity} = f'(t) = \lim_{h \rightarrow 0} \frac{f(t+h) - f(t)}{h}$$

This is the **rate of change of the position with respect to the time** at the time t and it is equal to the **slope of the tangent line at $(t, f(t))$** and it is called the

$f'(t) = \text{the derivative of } f \text{ at } t$

Derivative = limit of the slopes of the secant lines



Derivative = limit of the slopes of the secant lines

Go the Geogebra link to see the animation on [Secant Lines which tend to Tangent Lines](#):

[Secant Lines and Tangent Line](#)

Derivatives

Let $f : (b, c) \rightarrow \mathbb{R}$ be a function. The **derivative** of f at a is

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

This is the **rate of change of $f(x)$ with respect to x** and it is equal to the **slope of the tangent line at $(a, f(a))$** .

Equivalently, by the change of variables **$h = x - a$** , the **derivative of f at $x = a$** is

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

Tangent Line

Remark:

Compute a derivative means to solve an

indetermination of type $\frac{\rightarrow 0}{\rightarrow 0}$

Tangent Line

The equation of the **tangent line** to the curve $y = f(x)$ at the point $(a, f(a))$ is

$$L : y = f(a) + f'(a)(x - a)$$

Derivatives - Tangent Lines

Go the Wikipedia link to see the animation on [Tangent Lines](#):

[Tangent Lines](#).

Derivative: Leibnitz Notation

Other notation for the derivative of $f(x)$ is the following in the form

$$f'(x) = \frac{dy}{dx} = \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x}$$

Derivatives and Tangent Lines

- 1) Compute the derivative of $f(x) = c$ at any $x \in \mathbb{R}$.
- 2) Compute the derivative of $f(x) = x$ at any $x \in \mathbb{R}$.

Solution 1) The **derivative** of $f(x) = c$ at any x is

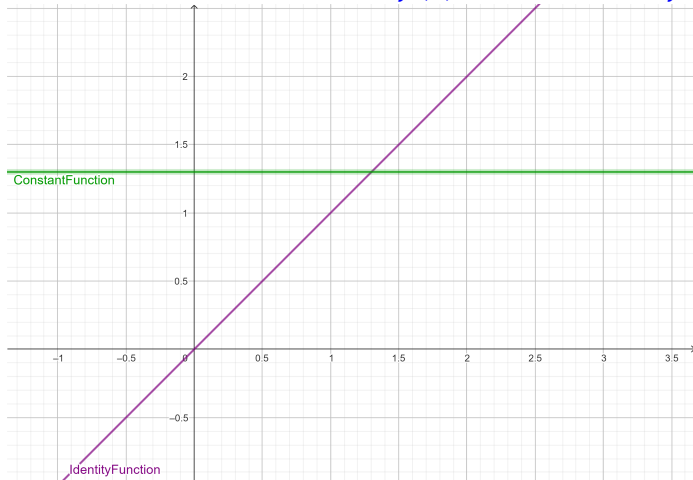
$$\boxed{f'(x) = c'} = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{c - c}{h} = \lim_{h \rightarrow 0} \frac{0}{h} \quad \boxed{= 0}$$

Solution 2) The **derivative** of $f(x) = x$ at any x is

$$\begin{aligned} \boxed{f'(x) = x'} &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{(x+h) - x}{h} \\ &= \lim_{h \rightarrow 0} \frac{h}{h} = \lim_{h \rightarrow 0} 1 = \boxed{= 1} \end{aligned}$$

Derivatives and Tangent Lines

- The slope of the tangent line of $f(x) = c$ at any x is $f'(x) = 0$



- The slope of the tangent line of $f(x) = x$ at any x is $f'(x) = 1$

Derivatives and Tangent Lines

- 1) Compute the derivative of $f(x) = x^2$ at any $x \in \mathbb{R}$.
- 2) Find the equation of the **tangent line** to the curve $y = x^2$ at the points $(1, 1)$ and at $(-3, 9)$.

Solution. The **derivative** of f at any x is

$$\begin{aligned} f'(x) &= (x^2)' = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{(x+h)^2 - x^2}{h} \\ &= \lim_{h \rightarrow 0} \frac{x^2 + 2xh + h^2 - x^2}{h} = \lim_{h \rightarrow 0} \frac{2xh + h^2}{h} = \lim_{h \rightarrow 0} \frac{h(2x + h)}{h} \\ &= \lim_{h \rightarrow 0} (2x + h) = 2x \end{aligned}$$

Tangent Lines

2) The equation of the **tangent line** at $(a, f(a))$ is

$$y = f(a) + f'(a)(x - a)$$

- Hence, at $(1, f(1))$ we have the slope $m = f'(1) = 2$ and $f(1) = 1$, so the equation of the tangent line is

$$L : y = 1 + 2(x - 1) \checkmark$$

- The equation of the **tangent line** at $(-3, f(-3)) = (-3, 9)$ is

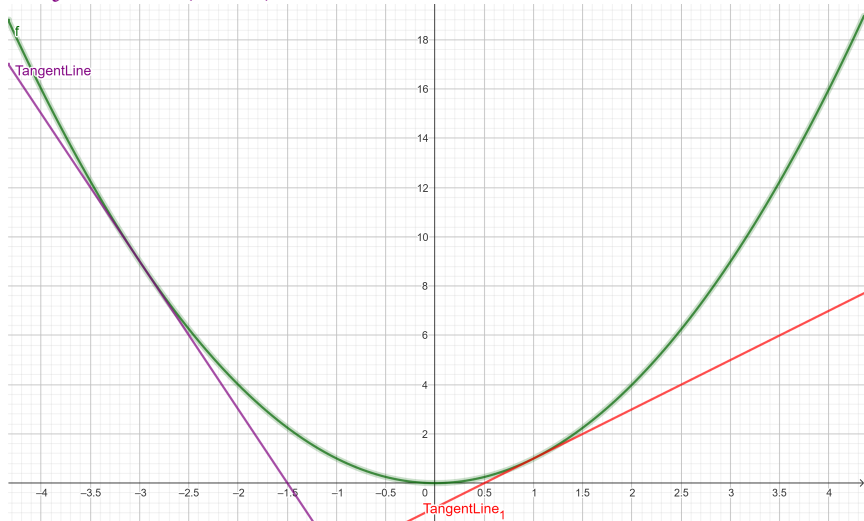
$$y = f(-3) + f'(-3)(x + 3)$$

Here, the slope is $m = f'(-3) = -6$ and $f(-3) = 9$, so

$$L : y = 9 - 6(x + 3) \checkmark$$

Tangent Lines for $f(x) = x^2$

$$L : y = 9 - 6(x + 3)$$



$$L : y = 1 + 2(x - 1)$$

Derivative of x^3

Compute the derivative of $f(x) = x^3$ at any $x \in \mathbb{R}$.

Solution. The **derivative** of $f(x) = x^3$ at any x is

$$\begin{aligned} f'(x) = (x^3)' &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{(x+h)^3 - x^3}{h} \\ &= \lim_{h \rightarrow 0} \frac{x^3 + 3x^2h + 3xh^2 + h^3 - x^3}{h} = \lim_{h \rightarrow 0} \frac{h(3x^2 + 3xh + h^2)}{h} \\ &= \lim_{h \rightarrow 0} (3x^2 + 3xh + h^2) = 3x^2 \checkmark \end{aligned}$$

Rules of Derivatives of the Main Functions

$$\text{If } f(x) = c \Rightarrow f'(x) = 0$$

$$\text{If } f(x) = x \Rightarrow f'(x) = 1$$

$$\text{If } f(x) = x^2 \Rightarrow f'(x) = 2x$$

$$\text{If } f(x) = x^3 \Rightarrow f'(x) = 3x^2$$

$$\text{If } n \text{ is an integer and } f(x) = x^n \Rightarrow f'(x) = nx^{n-1}$$

Derivative of $f(x) = \sqrt{x}$ only for $x > 0$

Compute the derivative of $f(x) = \sqrt{x}$ at any $x > 0$. Explain why f is not differentiable at $x = 0$.

Solution. The **derivative** of f at any $x > 0$ is

$$\begin{aligned} f'(x) &= (\sqrt{x})' = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{\sqrt{x+h} - \sqrt{x}}{h} \\ &= \lim_{h \rightarrow 0} \frac{(\sqrt{x+h} - \sqrt{x})(\sqrt{x+h} + \sqrt{x})}{h(\sqrt{x+h} + \sqrt{x})} = \lim_{h \rightarrow 0} \frac{x+h-x}{h(\sqrt{x+h} + \sqrt{x})} \\ &= \lim_{h \rightarrow 0} \frac{h}{h(\sqrt{x+h} + \sqrt{x})} = \lim_{h \rightarrow 0} \frac{1}{(\sqrt{x+h} + \sqrt{x})} = \frac{1}{2\sqrt{x}} \checkmark \end{aligned}$$

Derivative of $f(x) = \sqrt{x}$ doesn't exist at $x = 0$

f is not differentiable at $x = 0$.

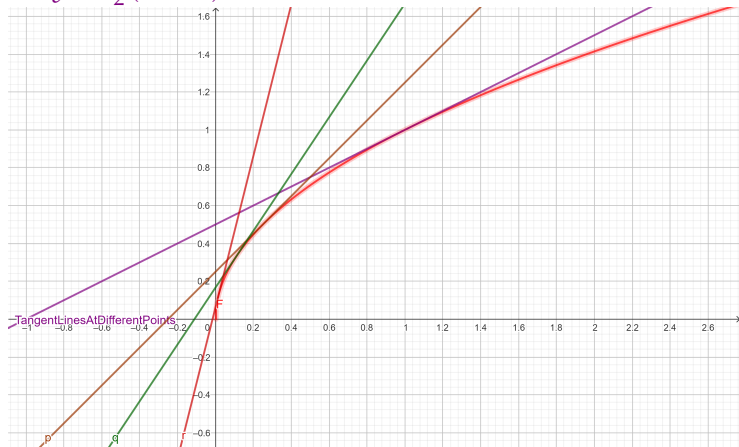
Solution. In the computation for the **derivative** of f at $x = 0$, it is only possible to approach $h \rightarrow 0$ from the right, so

$$\begin{aligned} f'(0) &= \lim_{h \rightarrow 0^+} \frac{f(0+h) - f(0)}{h} = \lim_{h \rightarrow 0^+} \frac{\sqrt{0+h} - \sqrt{0}}{h} \\ &= \lim_{h \rightarrow 0^+} \frac{\sqrt{h}}{h} = \lim_{h \rightarrow 0^+} \frac{1}{\sqrt{h}} = +\infty \end{aligned}$$

$\Rightarrow f'(0)$ doesn't exist ✓

Tangent Lines for $f(x) = \sqrt{x}$

$$L : y = \frac{1}{2}(x - 1) + 1$$



$$L : y = 4\left(x - \frac{1}{64}\right) + \frac{1}{8}$$

Rules of Derivatives of the Main Functions

- If $n > 0$ is an integer, $x \neq 0$, x in the domain of f and $f(x) = x^{\frac{1}{n}} = \sqrt[n]{x}$

$$\Rightarrow f'(x) = \frac{1}{n} \cdot \frac{1}{x^{1-1/n}}$$

- For example, with $r = \frac{1}{2}$:

$$\text{If } f(x) = \sqrt{x} \Rightarrow f'(x) = \frac{1}{2\sqrt{x}} \text{ only for } x > 0$$

- For example, with $r = \frac{1}{3}$:

$$\text{If } f(x) = \sqrt[3]{x} \Rightarrow f'(x) = \frac{1}{3 x^{2/3}} = \frac{1}{3\sqrt[3]{x^2}} \text{ for } x \neq 0$$

Rules of Derivatives of the rational powers

For $x \neq 0$, $x \in \text{Dom}(f)$, $r \neq 0$ a rational number $f(x) = x^r$, then its derivative is

$$f'(x) = r \cdot x^{r-1}$$

Notice: If $r = \frac{1}{n}$ and n is even, then the domain of the derivative is only $\text{Dom}(f') = (0, \infty)$:

- For example, with $r = \frac{1}{4}$:

$$\text{If } f(x) = x^{1/4} = \sqrt[4]{x} \Rightarrow f'(x) = \frac{1}{4 x^{3/4}} \text{ for } x > 0$$

Differentiable Functions

- A function f is differentiable at $x = a$ if $f'(a)$ exists.
- f is differentiable on an open interval

$$(a, b), \quad (a, \infty), \quad (-\infty, b), \quad (-\infty, \infty)$$

if it is differentiable at every number in the interval.

Rules of Derivatives of the Main Functions

The following functions are **differentiable** in their respective domains, unless an additional restriction is specified:

$f(x)$	$f'(x)$
c	0
x	1
x^2	$2x$
x^3	$3x^2$
x^n	nx^{n-1}
$\sqrt{x}$	$\frac{1}{2\sqrt{x}}$ only for $x > 0$
x^r	rx^{r-1} only for $x \in \text{Dom}(f), x \neq 0$
a^x	$\ln(a)a^x$
e^x	e^x
$\sin x$	$\cos x$
$\cos x$	$-\sin x$
$\ln(x)$	$\frac{1}{x}$

Rules of Derivatives of the Main Functions

The following functions are **differentiable** in their respective domains, unless an additional restriction is specified:

$f(x)$	$f'(x)$
a^x	$\ln(a)a^x$
e^x	e^x
$\log_b(x)$	$\frac{1}{\ln(b)x}$ only for $x > 0$
$\ln(x)$	$\frac{1}{x}$ only for $x > 0$

Differentiation Formulas

Let f, g be two differentiable functions, then the following are also differentiable and the derivative of the new function is obtained from the derivatives of f and g :

- $(f + g)' = f' + g'$

- $(f - g)' = f' - g'$

- $(cf)' = cf'$

- $(fg)' = f'g + fg'$

- $\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$

Examples: find the derivative

$$\text{a) } f(x) = 2\pi + \sin(x) \cdot e^x$$

$$\text{b) } g(x) = \frac{\sin(x) \cdot e^x}{(x^{1/3} + 2)}.$$

Solution. a)

$$f(x) = 2\pi + \sin(x) \cdot e^x \Rightarrow f'(x) = \cos(x) \cdot e^x + \sin(x) \cdot e^x \checkmark.$$

$$\begin{aligned} \text{b) } g(x) &= \frac{h(x)}{(x^{1/3} + 2)} \Rightarrow g'(x) = \frac{h'(x)(x^{1/3} + 2) - h(x)\frac{1}{3}x^{-2/3}}{(x^{1/3} + 2)^2} \\ &= \frac{(\cos(x) \cdot e^x + \sin(x) \cdot e^x)(x^{1/3} + 2) - \sin(x) \cdot e^x \frac{1}{3}x^{-2/3}}{(x^{1/3} + 2)^2} \\ &= \frac{(\cos(x) + \sin(x)) \cdot e^x}{(x^{1/3} + 2)} - \frac{\sin(x) \cdot e^x}{3x^{2/3}(x^{1/3} + 2)^2} \text{ only } \forall x \neq 0 \checkmark \end{aligned}$$

The Tangent Line of any linear function is the same line at any x

If $f(x) = mx + b$ then its derivative is, according to the rules,

$$f'(x) = m \cdot (x)' + (b)' = m$$

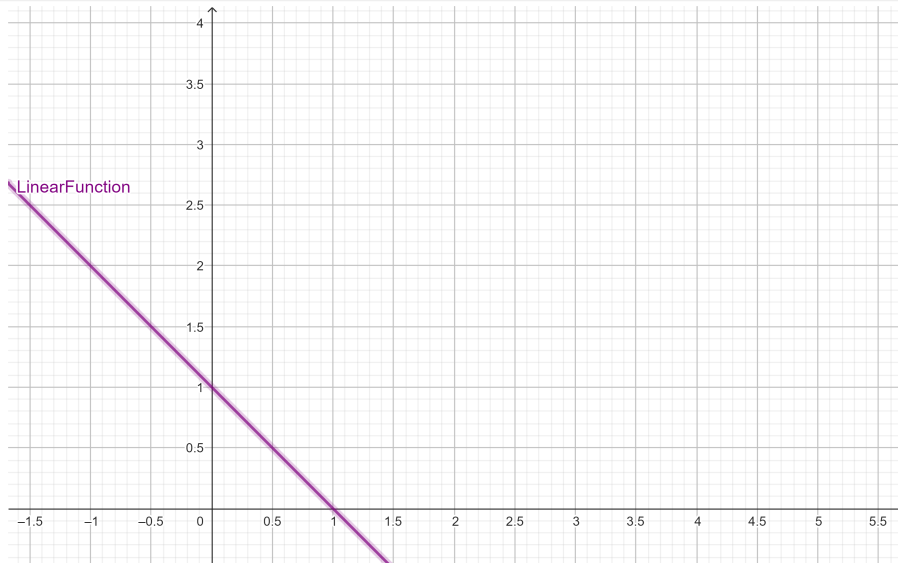
Hence, the tangent line at any point $(a, f(a)) = (a, ma + b)$ is

$$L : y = m(x - a) + f(a)$$

so, it is $L : y = m(x - a) + ma + b$ then

$$L : y = mx + b \checkmark$$

The Tangent Line of any linear function is the same line at any x



Derivatives and Tangent Lines

- 1) Compute the derivative of $f(x) = e^x + 2 \cos x + 4x^3$ at any $x \in \mathbb{R}$.
- 2) Find the equation of the **tangent line** to the curve $y = f(x)$ at the point $(0, 3)$.

Solution. 1) The **derivative** of f at any x is

$$f'(x) = (e^x)' + 2(\cos x)' + 4(x^3)' = e^x - 2 \sin x + 4 \cdot 3x^2$$

2) The equation of the **tangent line** at $(a, f(a))$ is

$$y = f(a) + f'(a)(x - a)$$

- Hence, at $(0, f(0)) = (0, 3)$ and the slope is $m = f'(0) = e^0 - 2 \sin(0) = 1$, so the equation of the tangent line is

$$L : y = 3 + x \quad \checkmark$$

Tangent line at $(0, 3)$: $L : y = 3 + x$ ✓



Find the intervals where $f(x) = |x|$ is differentiable

The question is for what values of x the derivative of $f(x) = |x|$ exists:

Solution. The derivative of $f(x) = |x|$ at any x is

$$f'(x) = |x|' = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{|x+h| - |x|}{h}$$

To continue, we need to separate in cases, since $|x|$ is a piece-wise function:

- For $x > 0 \Rightarrow |x| = x$, then

$$\begin{aligned} f'(x) = |x|' &= \lim_{h \rightarrow 0} \frac{|x+h| - |x|}{h} = \lim_{h \rightarrow 0} \frac{(x+h) - x}{h} \\ &= \lim_{h \rightarrow 0} \frac{h}{h} = \lim_{h \rightarrow 0} 1 = 1 \end{aligned}$$

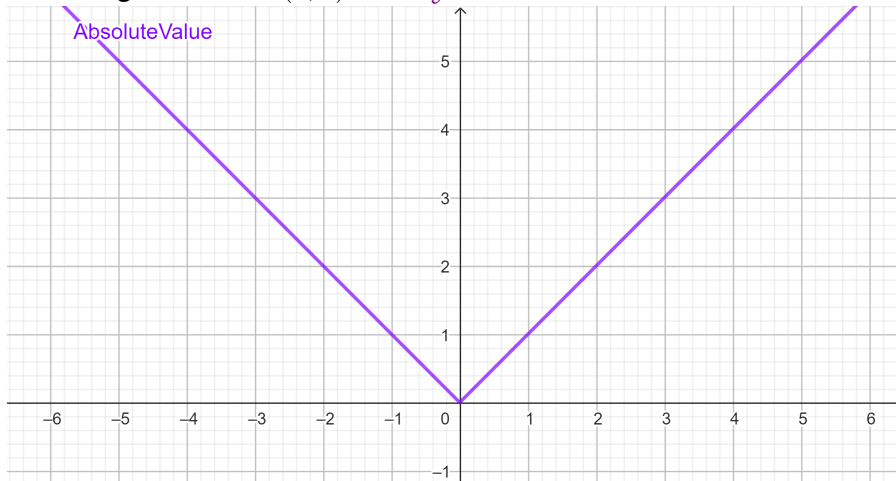
Derivative of $f(x) = |x|$ for $x < 0$

- For $x < 0 \Rightarrow |x| = -x$, then, if h is small enough, also $x + h < 0 \Rightarrow |x + h| = -(x + h)$:

$$\begin{aligned} f'(x) = |x|' &= \lim_{h \rightarrow 0} \frac{|x + h| - |x|}{h} = \lim_{h \rightarrow 0} \frac{-(x + h) - (-x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{-x - h + x}{h} \\ &= \lim_{h \rightarrow 0} \frac{-h}{h} = \lim_{h \rightarrow 0} -1 = -1 \end{aligned}$$

Tangent Lines for $f(x) = |x|$

The tangent line at (a, a) is $L : y = x \ \forall a > 0$



The tangent line at $(a, -a)$ is $L : y = -x \ \forall a < 0$

Derivative of $f(x) = |x|$ for $x = 0$ doesn't exist

- For $x = 0$, the derivative **doesn't exist**:

$$\boxed{f'(0)} = \lim_{h \rightarrow 0} \frac{|0 + h| - |0|}{h} = \lim_{h \rightarrow 0} \frac{|h|}{h} = \begin{cases} \lim_{h \rightarrow 0^+} \frac{h}{h} & \text{since } h > 0 \\ \lim_{h \rightarrow 0^-} \frac{-h}{h} & \text{since } h < 0 \end{cases}$$

$$= \begin{cases} 1 & \text{as } h \rightarrow 0^+ \\ -1 & \text{as } h \rightarrow 0^- \end{cases} \Rightarrow \text{the limit doesn't exist}$$

$\Rightarrow$ f is not differentiable at $x = 0$.

In fact, the geometrical quality of $|x|$ having **a corner** at $(0, 0)$ implies that f has not a tangent line at $(0, 0)$.

$$|x|' = \begin{cases} 1 & \text{in } (0, \infty) \\ -1 & \text{in } (-\infty, 0) \end{cases} \checkmark$$

Differentiability implies Continuity

Theorem

If f is differentiable at a , then f is continuous at a .

- The reciprocal is not true: **There are continuous functions which are not differentiable:**

$f(x) = |x|$ is continuous but not differentiable at $x = 0$.

Differentiability implies Continuity

Proof. In order to proof that f is continuous at $x = a$, we need to show that

$$\lim_{x \rightarrow a} f(x) = f(a)$$

- Let f be differentiable at $a \Rightarrow x = a \in \text{Dom}(f)$ and there exists

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{(x - a)}.$$

- Using the product law, we can multiply the previous limit by the factor $\lim_{x \rightarrow a} (x - a) = 0$, so the result is zero:

$$\begin{aligned} 0 = f'(a) \cdot 0 &= \lim_{x \rightarrow a} \frac{f(x) - f(a)}{(x - a)} \lim_{x \rightarrow a} (x - a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{(x - a)} \cdot (x - a) \\ &= \lim_{x \rightarrow a} [f(x) - f(a)] \end{aligned}$$

Hence, $0 = \lim_{x \rightarrow a} [f(x) - f(a)]$, which means that

$$f(a) = \lim_{x \rightarrow a} f(x) \quad \checkmark$$

Three ways for f not to be differentiable at a

f is differentiable at $a \in \text{Dom}(f)$ if and only if the limit of the quotient exists:

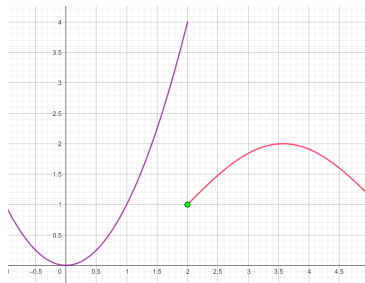
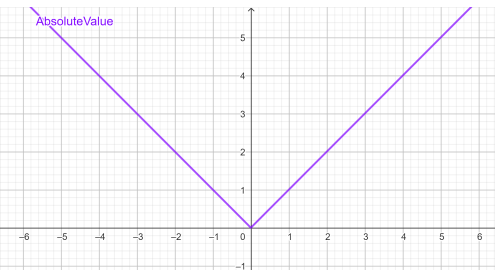
$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{(x-a)}$$

But the limit might not exist in different ways:

- If the right and left side limits are different: f has a corner
- If f is not continuous with a jump discontinuity: the limit of the quotient exists from the right but not from the left (or from the left but not from the right).
- If the limit of the quotient is infinity: the tangent line would be vertical, but it is not accepted as tangent line.

Three geometric ways for f not to be differentiable at a

$f(x) = |x|$ with a corner

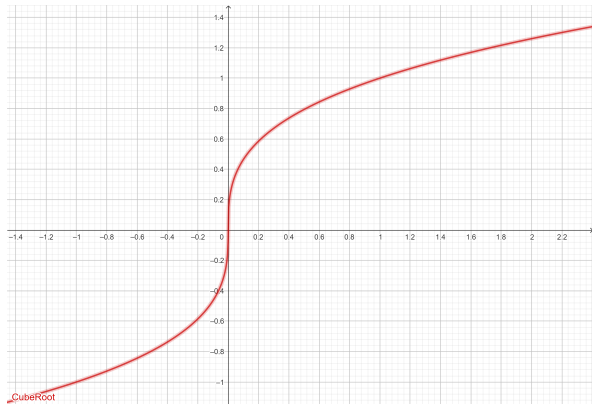


$g(x)$ with jump discontinuity

Three geometric ways for f not to be differentiable at a

$h(x) = x^{1/3}$: the derivative does not exist at $x = 0$ since

$$\nexists \lim_{h \rightarrow 0} \frac{f(0+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{\sqrt[3]{h} - 0}{h} = \lim_{h \rightarrow 0} \frac{1}{\sqrt[3]{h^2}} = \infty$$



Example

Find all $x \in \mathbb{R}$ the tangent lines of f and g are parallel:

$$f(x) = x^3 + 3x^2 + 3x + 4, \quad g(x) = x^3 - \frac{3}{2}x^2 - 6x + 1$$

and find the equation of the lines at each of those points.

Solution: Compute the derivatives of both functions:

$$3x^2 + 6x + 3 = f'(x) = g'(x) = 3x^2 - 3x - 6$$

Then, let us find x such that the equality holds:

$$3x^2 + 6x + 3 = 3x^2 - 3x - 6 \iff 0 = 9x + 9 = 9(x + 1)$$

Then, the only possibility is

$$x = -1$$

Hence, we conclude that the tangent lines of f and g are parallel at $x = -1$ and the equations are as follows:

$$y = f(a) + f'(a)(x - a)$$

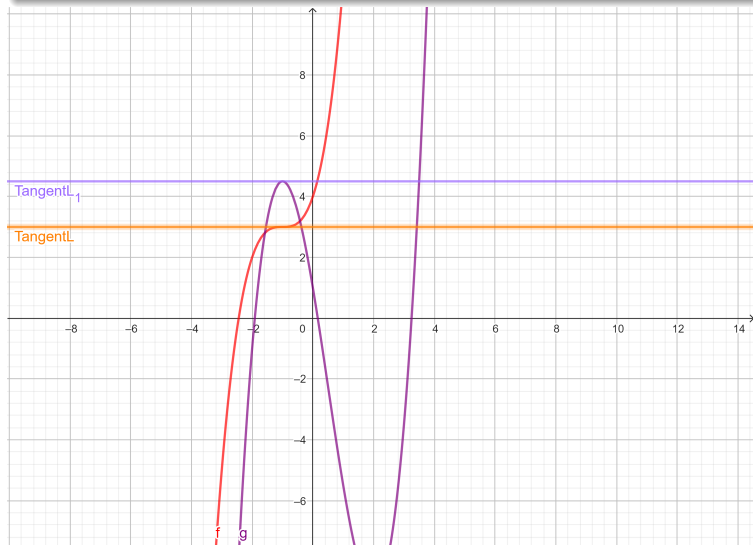
- Tangent line of f at $(-1, f(-1)) = (-1, 3)$ with slope $m = f'(-1) = 0$:

$$y = 3$$

- Tangent line of f at $(-1, g(-1)) = (-1, \frac{9}{2})$ with slope $m = g'(-1) = 0$:

$$y = \frac{9}{2}$$

The tangent lines of f and g are parallel at $x = -1$



$$L_f : y = 3$$

$$L_g : y = \frac{9}{2}$$

Higher Order Derivatives

Given a differentiable function f for which also f' is also differentiable, we may consider its derivative:

$$f''(x) = \frac{d}{dx} f'(x) = \frac{d}{dx} \left(\frac{d}{dx} f \right) (x)$$

- A function can have **higher order derivatives** $f', f'', f''', f^{(4)}$, and so on, each of them exists whenever the previous one is differentiable.

Ex: $f(x) = 7x^3 \Rightarrow f'(x) = 7 \cdot 3x^2 \Rightarrow f''(x) = 7 \cdot 3 \cdot 2x \Rightarrow f'''(x) = 42 \Rightarrow f^{(4)}(x) = 0$ ✓.

Then, the **higher order derivatives** in this case $f^{(4)}, f^{(5)}$, and $f^{(n)}$ are all zero, for any $n \geq 4$ ✓.

Higher Order Derivatives

- If $f(x)$ is the position of an object at the time x
- f' is the velocity at the time x , it is the rate of change of the position respect to the time.
- f'' is the acceleration at the time x , it is the rate of change of the slope of the original curve.
- Thus the acceleration function is the derivative of the velocity function and is therefore the **second derivative of the position function**:
Acceleration it is the change in velocity.

Ex. 6. If $f : (0, \frac{\pi}{2}) \rightarrow \mathbb{R}$, find the coordinates of $(a, f(a))$ such that the tangent line to the curve $y = f(x)$ is parallel to the line L: $y = 3$:

$$f(x) = 3 + \frac{\sin(x)}{(2 - \sqrt{2} \cos(x))}$$